More Function Exercises: Periodicity, Piecewise Definitions, Images, and Functional Equations

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§1.1 Piecewise Maps and Their Domains

Function exercises of this kind are solved by translating each problem into one of several languages:

formula, graph, image/preimage, iteration, functional equation.

Many errors come from using a formula while ignoring its domain or treating an identity valid for all inputs as though it were a formula at a few special values. We follow the standard real-analysis approach to continuous images and the Cauchy equation, as in Gressman's *Advanced Analysis* notes and the lecture notes on Cauchy's functional equation.[1]

A piecewise-defined function may use different formulas on intervals such as

$$x < 0, \qquad x = 0, \qquad x > 0.$$

When studying continuity or solving equations involving such a function, one must first decide which case the input belongs to. If the input itself is an expression such as $f(x)$ or $x + 1$, the relevant case may change during the calculation.

The safe method is:

(i) identify the possible regions of the input;

(ii) solve the equation in each region;

(iii) check that the solution actually belongs to the assumed region.

The pages include small sketches of graphs. The standard transformations are:

$$y = f(x - a) \quad \text{shift right by } a,$$

$$y = f(x) + b \quad \text{shift up by } b,$$

$$y = -f(x) \quad \text{reflect across the } x\text{-axis},$$

$$y = f(-x) \quad \text{reflect across the } y\text{-axis}.$$

Absolute values fold graphs:

$$y = |f(x)|$$

reflects the negative part upward, while

$$y = f(|x|)$$

copies the right half to the left.

[1] P. Gressman's intermediate-value theorem notes, with complementary calculus exercises from Carleton University.

A piecewise function is a function defined after partitioning the domain into regions. The mathematical issue is not only which formula applies, but how the pieces meet. Discontinuities occur at boundaries of the partition; differentiability requires matching first-order behavior across boundaries.

For example, if

$$f(x) = \begin{cases} f_1(x), & x < a, \\ f_2(x), & x \geq a, \end{cases}$$

then continuity at a requires

$$\lim_{x \to a^-} f_1(x) = f_2(a).$$

Differentiability additionally requires matching slopes. This is a local gluing condition. For example, let $f(x) = ax^2$ for $x < 1$ and $f(x) = bx + c$ for $x \geq 1$. Continuity at 1 requires $a = b + c$, while differentiability requires in addition $2a = b$. Hence differentiability holds exactly when $b = 2a$ and $c = -a$. The boundary test is an actual linear system for the unknown coefficients, rather than an informal statement that the formulas should match.

§1.2 Images of intervals and finite maps

For a continuous function on an interval, the image is also an interval. Indeed, intervals are the connected subsets of $\mathbb{R}$, and continuous images of connected sets remain connected. This is the intermediate value theorem in set language. However, the image of an open interval need not be open: $f(x) = x^2$ sends $(-1, 1)$ to $[0, 1)$. If the function is monotone on the interval, then the endpoints determine the image directly.

To find the range of an expression, rewrite it as a one-variable function, determine monotonicity or extrema, and translate the result back into an interval.

The printed exercise page also touches finite functions. If $|A| = m$ and $|B| = n$, then the number of functions $A \to B$ is

$$n^m,$$

because each element of A independently chooses one image in B.

If one requires injectivity and $m \leq n$, the number is

$$n(n-1)\cdots(n-m+1).$$

If one requires bijectivity and $m = n$, the number is

$$n!.$$

If surjectivity is required for $m \geq n$, inclusion–exclusion gives

$$\#\operatorname{Surj}(A, B) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} (n-j)^m.$$

Indeed, count all n^m functions and subtract those omitting at least one of the n target elements. This explains why injectivity and surjectivity lead to different counting problems.

§1.3 Functional equations as transformations

A relation such as

$$f(x+1) = 1 - f(x)$$

implies

$$f(x+2) = 1 - f(x+1) = f(x).$$

Thus the function is periodic with period 2. The general method is to apply the same equation repeatedly until the input returns to a familiar form.

If instead

$$f(x+T) = f(x)$$

for all x, then T is a period. The smallest positive period, if it exists, is called the fundamental period.

Substituting special values such as $x = 0$, $y = 0$, or $y = x$ is a standard strategy. For a functional equation in two variables,

$$F(x, y, f(x), f(y), f(x+y)) = 0,$$

choosing simple values isolates pieces of the function.

For example, if

$$f(x+y) = f(x) + f(y) + c,$$

then setting $x = y = 0$ determines $f(0)$, and subtracting a constant can reduce the equation to the additive Cauchy equation.

A functional equation often specifies a transformation rule rather than a direct formula. For example, equations involving $f(x+1)$, $f(2x)$, or $f(f(x))$ are naturally studied by iteration. The orbit of x under a transformation T is

$$x,\ T(x),\ T^2(x),\ \ldots.$$

A functional equation may prescribe how f behaves along these orbits.

For instance, if

$$f(x+1) = f(x),$$

then f is constant along orbits of the translation $T(x) = x + 1$. Periodicity is invariance under a group action of $\mathbb{Z}$.

A functional equation often says that f is invariant or equivariant under transformations. The equation

$$f(x+T) = f(x)$$

says f is invariant under the translation action of $T\mathbb{Z}$. The equation

$$f(ax) = bf(x)$$

says f is homogeneous with respect to scaling.

Thus solving a functional equation often means finding the invariants of a group or semigroup action. This perspective makes the original examples more natural: periodicity, parity, and scaling are all symmetry constraints.

§1.4 Regularity makes Cauchy's equation rigid

Cauchy's equation is

$$f(x+y) = f(x) + f(y).$$

For an additive map $f : \mathbb{R} \to \mathbb{R}$, continuity at one point, Lebesgue measurability, or boundedness on a nondegenerate interval each implies

$$f(x) = cx.$$

Here is the continuous case. Additivity gives $f(0) = 0$, $f(nx) = nf(x)$, and $f(p/q) = (p/q)f(1)$ for rationals p/q. If $q_k \in \mathbb{Q}$ approaches real x, continuity yields $f(x) = \lim_k f(q_k) = xf(1)$. The discontinuous Hamel-basis solutions show exactly what continuity rules out. But without regularity assumptions, there are wild solutions constructed using a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$.

This shows why elementary functional equations often include hidden conditions such as continuity or monotonicity. These conditions are not technical decorations; they exclude pathological solutions arising from the axiom of choice.

§1.5 From iterated intervals to symbolic dynamics

Piecewise-defined functions and iteration can lead to symbolic dynamics. If an interval map sends subintervals across one another, one can encode an orbit by a sequence of symbols recording which subinterval contains each iterate.

The shift map

$$\sigma((a_0, a_1, a_2, \ldots)) = (a_1, a_2, a_3, \ldots)$$

is a central model. Complicated interval dynamics can often be compared to shifts on symbol spaces.

For a specific example, partition $[0,1)$ into $I_0 = [0, 1/2)$ and $I_1 = [1/2, 1)$, and set $T(x) = 2x \pmod 1$. The itinerary of x is the sequence $a_k \in \{0,1\}$ recording which interval contains $T^k(x)$. Under the binary expansion convention that avoids trailing strings of ones, this itinerary is precisely the binary digits of x; the map T shifts them left. Thus symbolic dynamics is the exact encoding of a concrete interval map, with the usual care about dyadic endpoints.